

Batch: DMW 2 Roll No.: 16010324044
Experiment/ Assignment /Tutorial No.: 04
Grade: AA / AB / BB / BC / CC / CD /DD
Signature of the Staff In-charge with date

TITLE: K – nearest neighbour for classification and regression

AIM: a) To implement k- nearest neighbour algorithm for classification and regression

b) To perform k-fold cross validation to determine hyperparameters

c) To observe effect of k-value, distance measure, weights, search methods on classification and regression performance

OUTCOME: Students will be able to implement k nearest neighbour algorithm for classification and regression tasks

Procedure: I] K-Nearest Neighbour for Classification

1. Read Raisin data and identify predictor and response variables
2. Find number of data instances
3. Find the types of data variables
4. Split data into train and test sets, find dimension of train and test sets
5. Perform k-fold cross validation on train data using GridSearch method to find the best hyperparameters: value of k, weights , search method, distance measure. Note down the best hyperparameters obtained.
6. Train k- nearest neighbour classifier model using the best hyperparameters on train data and perform prediction on test data. Note performance metrics and comment on it.
7. Plot confusion matrix
8. Plot Receiver Operating Characteristic curve and note value of AreaUnderCurve. Comment on the result.
9. Repeat the above process for Pumpkin_Seeds.csv dataset and write a brief summary from your observations and results.
10. Apply MinMax scaling to train and test data and repeat above steps. Report your observations regarding classification performance on test data

Procedure: II] K-Nearest Neighbour for Regression

11. Read Advertising.csv data and identify predictor and response variables
12. Find number of data instances
13. Find the types of data variables
14. Split data into train and test sets, find dimension of train and test sets

15. Perform k-fold cross validation on train data using GridSearch method to find the best hyperparameters: value of k, weights, search method, distance measure. Note down the best hyperparameters obtained.
16. Train k- nearest neighbour regressor model using the best hyperparameters on train data and perform prediction on test data. Note performance metrics and comment on it.
17. Train k- nearest neighbour regressor model with default values and perform prediction on test data. Note performance metrics and compare with results obtained in step 16.

Optional task/Practice: Repeat Procedure I and II for “Raisins.csv” and “data_k-Barriers.csv” datasets respectively whose description is available in the uploaded research paper. Write a brief summary from your observations and results.

Observation Table:

Part A – Pumpkin Seed Dataset (KNN Classification)

Task 1: Dataset Understanding

Observe the dataset and answer:

How many records are present in the dataset?

How many input features are used for classification?

What is the target (class) variable?

How many classes are present in the dataset?

Task 2: Data Preprocessing

Observe the output and answer:

Are there any missing values?

Are there any duplicate records?

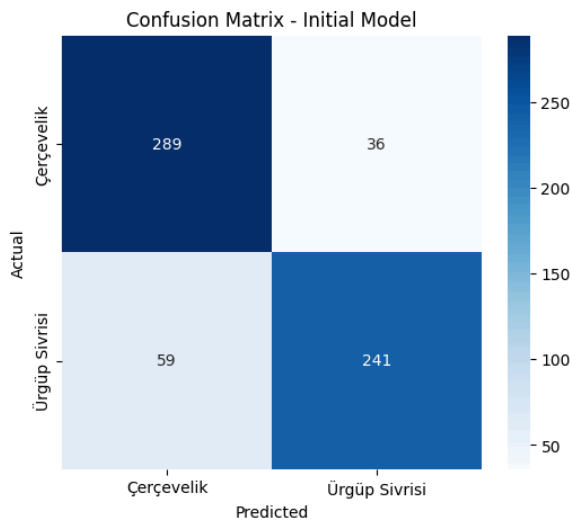
Why is Label Encoding applied?

Why is Feature Scaling necessary before applying KNN?

Task 3: Initial KNN Model

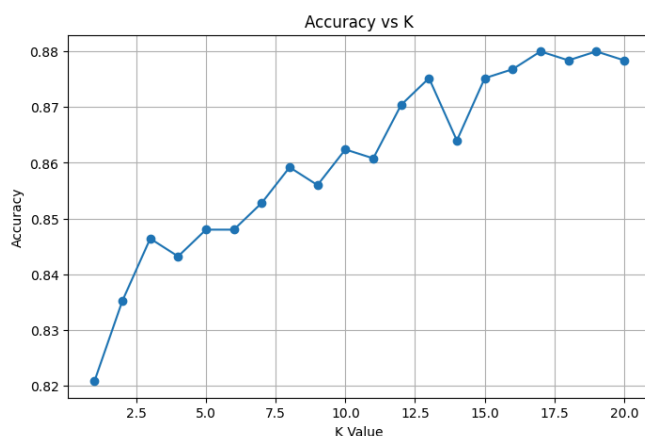
After training the model,

Parameter	Observation
Initial K value	5
Initial Accuracy	84.8
Number of Correct Predictions	530
Number of Incorrect Predictions	95



Observe the confusion matrix.

- Which class has the highest number of correct predictions?**
Çerçvelik is the class has the highest number of classifications
- Which class is most frequently misclassified?**
Ürgüp Sivrisi is the class which is most frequently misclassified
- What does each diagonal element represent?**
The diagonal elements represent the correct predictions. They show how many samples of each class were correctly classified by the KNN model.
- What do the off-diagonal elements indicate?**
The off-diagonal elements represent incorrect predictions or misclassifications. They show when one class is wrongly predicted as another class.
- Why are Precision and Recall important in classification?**
Precision tells us how many of the samples predicted as a particular class are actually from that class.



Recall tells us how many actual samples of a class were correctly identified.

Task 4: Accuracy vs K Graph

1. Which K value gives maximum accuracy?

17 is the k value which gives maximum accuracy

2. What happens when K becomes very large?

The accuracy will decrease as the model considers too many neighbours

3. Why doesn't K=1 always produce the best model?

Because it is sensitive to noise and individual data points

Task 5: Final Comparison

Parameter	Initial Model	Optimized Model
Parameter		
Accuracy		
Best K		

1. Did optimization improve performance?

2. By approximately how much did the accuracy improve?

3. Which model would you recommend for deployment? Why?

Part B – Advertisement Dataset (KNN Regression)

Task 1: Dataset Understanding

1. Number of records?

200

2. Number of input variables?

3 (TV, Radio, Newspaper)

3. Target variable?

Sales

4. Is this a classification or regression problem? Justify.

Regression, since the data is numerical and continuous

Task 2: Initial KNN Regressor

Metric	Observation
Initial K	6
MAE	1.3568
RMSE	1.6858
R ² Score	0.8955

1. Is the prediction error high or low?

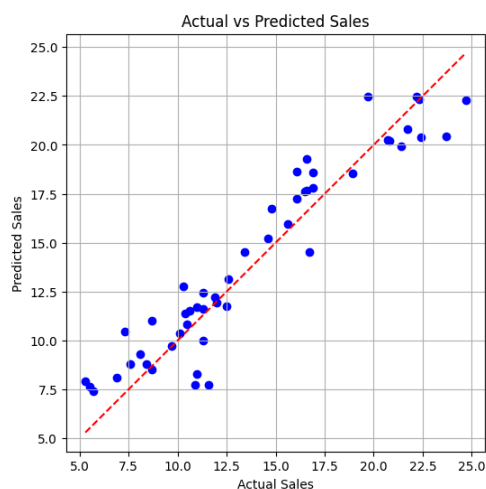
Low

2. What does the R² score indicate?

It indicates how well the regression line fits

3. Which metric is more useful for measuring prediction error—MAE or RMSE? Why?

RMSE, it is less sensitive to outliers



Task 3: Actual vs Predicted Scatter Plot

Observe the scatter plot.

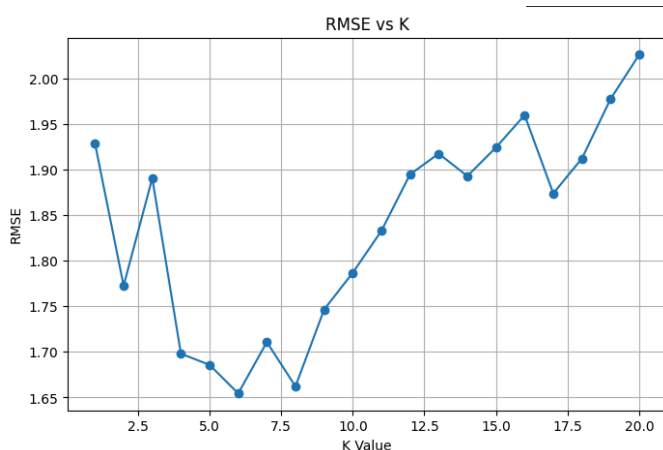
1. Are most points close to the reference line?

Almost

2. What does this indicate about prediction accuracy?

The accuracy is good enough and the predicted values will be closer to the actual

3. What would the graph look like if the model performed poorly?



There would be few points that lie or be close enough to the predicted line

Task 4: RMSE vs K Graph

1. At which K is RMSE minimum?

at K = 6

2. Why should we select the model with minimum RMSE?

Min RMSE, will result in the prediction value to be close enough to reality

3. What happens if K becomes too large?

Task 5: Final Comparison

Metric	Initial	Optimized
MAE	1.3568	1.2386
RMSE	1.6858	1.6549
R ² Score	0.8955	0.8993

1. Which model performs better?

Optimized

2. Did RMSE decrease after optimization?

Yes

3. Did the R^2 score improve?

Yes

4. Which model would you recommend? Why?

Optimized

Final Comparative Analysis

Parameter	Pumpkin Dataset	Advertisement Dataset
Machine Learning Task	Classification	Regression
Algorithm Used	KNeighborsClassifier	KNeighborsRegressor
Target Variable Type	Categorical / Discrete (Class)	Continuous / Numerical (Sales)
Evaluation Metrics	Accuracy, Precision, Recall, Confusion Matrix	MAE, MSE, RMSE, R^2 Score
Performance Measure	Classification Accuracy (%)	R^2 Score & RMSE
Hyperparameter Tuned	n_neighbors, weights, metric	n_neighbors, weights, metric
Final Best K	K = 17	K = 6

Part C – Dataset Analysis (Without Implementation)

Open the **Raisin Dataset** and **K-Barriers Dataset** and study their attributes. Based on observations, answer the following questions.

Question	Raisin Dataset	K-Barriers Dataset
Number of records	900	182
Number of input features	7	4
Target (output) attribute	1	1
Is the target variable categorical or numerical?	Categorical	Numerical
Which machine learning task is suitable? (Classification/Regression)	Classification	Regression

Question	Raisin Dataset	K-Barriers Dataset
Which KNN algorithm will be used? (Classifier/Regressor)	Classifier	Regressor
Which evaluation metrics are appropriate?	Confusion Matrix (Accuracy)	RMSE (Error)

Conclusion:

In this experiment, I demonstrated KNN can be effectively used for both classification and regression. Hyperparameter tuning improves model performance by selecting a suitable K-value, distance metric, and weighting method. Feature scaling is important because KNN is distance-based. Overall, the optimized models provided better accuracy for classification and lower prediction error for regression.

Signature of faculty in-charge with date